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Published in:
International Wound Journal

DOI:
10.1111/iwj.14026

Publication date:
2023

Document version:
Final published version

Document license:
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Citation for pulished version (APA):
Sondergaard, S. F., Vestergaard, E. G., Andersen, A. B., Kolbaek, R., Dahl, M., & Hogh, A. (2023). How patients with diabetic foot ulcers experience telemedicine solutions: A scoping review. *International Wound Journal*, *20*(5), 1796-1810. https://doi.org/10.1111/iwj.14026

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Download date: 07. Nov. 2026

Received: 19 October 2022 | Accepted: 14 November 2022

DOI: 10.1111/iwj.14026

REVIEW ARTICLE

IWJ WILEY

How patients with diabetic foot ulcers experience telemedicine solutions: A scoping review

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Abstract

Diabetic foot ulcer (DFU) is a common, complex and severe complication of diabetes that is associated with severely decreased health-related quality of life. Treatment of DFUs calls for a multi-sectoral approach, incorporating interdisciplinary care pathways. Telemedicine (TM) may be used as a communication tool between caregivers across healthcare sectors to obligate the need for close follow-up, including early intervention in preventing the recurrence of DFU. The objective of this review was to identify, examine and conceptually map the available literature on patients' experiences and views regarding the use of TM solutions among patients with DFUs. We identified the Population, Concept and Context to pinpoint the focus of this review, word the research question and title as well as facilitate the literature search strategy. The literature examined stems from 13 sources. We imposed no restrictions on the methodological approach of the included studies, neither on the format. During the review process, four main maps emerged: "A whole human not merely a hole in a human," "Less of a burden on the family, the community and the environment," "Competences and continuity of care are essential for high-quality care" and "The quality and modality of the technology." Further investigation from both the patients' and the multi-sectoral caregivers' perspective is needed, focusing on whatever modifications of the TM intervention may fit the DFU care pathway better.

KEYWORDS

diabetic foot ulcer, patient experience, scoping review, telemedicine, wound

Key Messages

- a TM solution was reported to promote holistic and person-centred care and treatment
- TM gave the patients an experience of being viewed as a whole person and not merely as a hole in a person
- using a TM solution promotes an experience of having control, being included in decision-making, and being able to lead an almost normal life

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- TM facilitates a feeling of empowerment, being less of a burden on the families, the community and the environment
- a poorly functioning solution could be a game changer for patients' views on the use of a TM solution

1 | BACKGROUND

Treatment of diabetes-related foot ulcers (DFU) is challenging because of a multifactorial ethology that often calls for multi-sectoral input with interdisciplinary foot teams.[1] DFUs are frequent and dangerous with an estimated lifetime prevalence of 19–24% combined with a high recurrence rate (40% in the first year and 65% in the first 3 years after healing)[1] and a 5-year shorter life expectancy when compared to patients only suffering from DM.[2-4] Furthermore, the prevalence of patients with DFUs is expected to rise over the next decades based on a general increase in life expectancy.[5-7]

In addition, treatment of patients with DFUs is often intensive, including frequently prolonged hospitalisation, which induces high costs reaching an expense level comparable to cancer treatment.[2,8] Diabetic complications in general are associated with severely decreased health-related quality of life and depression.[9] These findings are retrieved among people with DFUs combined with social isolation due to immobility as well as feelings insecurity and anxiety about the future.[4,10,11] The scenario is described as having a major impact on the patient's experiences of quality of life.[2,9,12]

In many countries, the use of different modalities of telemedicine (TM) is already implemented as a communication tool between caregivers across the health care sectors.[13]

According to Hazenberg et al, the objective of TM solutions is to improve the efficiency and effectiveness of care as well as patients' well-being and autonomy in a world with swiftly changing socio-economic perspectives in health care.[14] However, the TM modalities are developed and utilised differently on national levels.[9,13,15]

The use of TM for patients with DFUs opens up for multi-sectoral and interdisciplinary close follow-up, including monitoring and managing the care pathway as well as intervening early to prevent the recurrence of DFUs. The direct consequences of implementing TM for this patient group are described as (1) Optimising the cost-effectiveness of substituting a physical visit with a digitally supported appointment (2) Reducing the travel time for the patient while still providing a highly professional service.[16] However, no Patient Reported Outcome Measurements (PROMs) have been found more appropriate than others in assessing health-related quality of life among patients living with DFUs.[17,18] Thus, most research on TM for patients with diabetic foot ulcers focuses on diagnosis and self-monitoring, and is related to therapeutic purposes.[11,13]

Thus, the patient's perspective is underexplored however important for various reasons. Firstly, it is likely that the patient's experiences of TM are related to whether the intervention is cost-effective (since the patients must actively engage in using the equipment). Secondly, the patients' and their families' attitudes towards TM will probably affect the treatment outcomes and finally, TM solutions might also be experienced as important to improved quality of care, and thereby enhanced social support. Zhao et al argue that efforts to maintain general health and early intervention in preventing the occurrence of DFUs may help avoid the deterioration of physical health and health-related quality of life among this patient group.[11]

To our knowledge, only a limited number of published studies address the patient's experience and perspective using TM solutions in relation to the care and treatment of DFUs. The objective of this review was therefore to identify, examine and conceptually map the available literature on patients' experiences using TM solutions for remote monitoring, consulting and exchange of information between the patient, the community care service and the hospital specialists regarding the treatment and care of their DFUs.

2 | DESIGN AND METHOD

We followed the Joanna Briggs Institute methodological guidance for reporting a scoping review.[19] In addition, the scoping review was written in accordance with the PRISMA-ScR statement and checklist, in tandem with the JBI guidance.[19-21] Furthermore, a research librarian supervised our search process.

We used the mnemonic PCC (Population, Concept and Context) to identify the focus and context of the review, word the research question and title, develop the inclusion and exclusion criteria and facilitate the strategy for the literature search.[19] Table 1 summarises the PCC.

2.1 | Population

The sources were adult patients' (>18 years) experiences and views on the treatment and care of their DFUs.

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TABLE 1 Summarised PCC

Population	Concept	Context
Patients' experiences with the treatment and care of their diabetic foot ulcers.	Telemedicine technologies /modalities	Primary and secondary health care professionals remotely monitoring, consulting with each other and exchanging information

2.2 | Concept

We rely on the World Health Organisation (WHO) Global Observatory for e-Health's definition of TM, which is defined below:

> "The delivery of health care services, where distance is a critical factor, by all health care professionals using information and communication technologies for the exchange of valid information for the diagnosis, treatment, and prevention of disease and injuries, research and evaluation, and for the continuing education of health care providers, all in the interests of advancing the health of individuals and their communities."[22]

According to the WHO statement, TM may also include monitoring and storing information from patients.[22] The TM modalities we included were various kinds of smartphones, tablets, computers, computer software, smartphone/tablet apps and web cameras.

2.3 | Context

The context concerns the remote monitoring process, consultation and exchange of information between the primary and secondary health care sectors in relation to the treatment and care of patients with DFUs, undertaken in collaboration with the patient, potentially a wound care specialist at a hospital and the health professionals in the community care services.

3 | SEARCH STRATEGY

We aimed to construct a conceptual map by examining various types of literature, including scientific and grey sources.[19,21] We used the available literature on patients' experiences and views on using various modalities of TM

TABLE 2 Inclusion and exclusion criteria

Inclusion criteria	Exclusion criteria
Scientific and grey sources reporting on: • How patients with diabetic foot ulcers experience telemedicine solutions • Adult patient, age ≥ 18 years • Different telemedicine modalities used for the remote monitoring, consultation and exchange of information performed by the homecare professionals and hospital specialists for and with the patient Primary studies with no restrictions on methodological approach	Sources reporting on: • Telemedicine technologies used during admission to hospital • Telemedicine technologies used internally in primary care • Public social media used as a telemedicine solution Literature reviews Sources written in languages other than English, Danish, Swedish and Norwegian

(eg, the remote monitoring, consultation and exchange of information performed by the homecare professionals and hospital specialists for and with the patient) regarding the treatment and care of their DFUs. We imposed no restrictions on the methodological approach of the studies included (embracing both quantitative and qualitative data, mixed methodologies as well as experimental and intervention study designs whether randomised or non-randomised) or on the format (including reviews of various types, individual case reports, discussion papers and conference papers). In addition, studies in the Scandinavian languages (Danish, Swedish and Norwegian) and English were included. Table 2 presents the inclusion and exclusion criteria applied.

We followed a three-phased search strategy as described by Peters et al.[19] The first phase was a limited search in CINAHL, PubMed, Google, Google Scholar and Bibliotek.dk (Danish national public library portal). This phase aimed to get an overall view of the sources on the topic and to identify keywords and index terms that describe the topic of the scoping review. In the second phase, a systematic search in the databases PubMed, CINAHL, Web of Science and Embase databases was performed to identify scientific sources. The search was constructed using index words, such as MeSH terms, building blocks and the Boolean phrases AND and OR to systematise, broaden and subsequently decrease the number of hits in the search. Using the index phrase "*patient experience*," we reduced the number of hits to less than ten. Therefore, we extended the search without this search word to prevent excluding relevant sources.

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TABLE 3 Building block presentation from the search in the PubMed database

Building blocks			
---------- AND ----------			
OR	Patient″ [MeSH]	Diabetic Foot″ [MeSH]	Telemedicine″ [MeSH]
		• Diabet* wound*	• Telehealth*
		• Diabet* ulcer*	• Mobile Health*
		• Diabet* foot	• Mhealth*
		• Diabet* feet	• Ehealth*
		• Foot ulcer*	• Remote consultation*
		• Wound care	• Telecommunication*
		• Wound management	• Web-based consultation*
			• Internet-based consultation*
			• Internet-based counselling*

In the third phase, we reviewed the reference lists from the literature identified in order to find unidentified references from our systematic searches. As an example, Table 3 illustrates the building blocks with examples from the PubMed database search.

To identify sources in the grey literature, we performed searches on Google.dk, Google Scholar and Open Grey. Finally, we hand-searched websites and reports from different organisations in Denmark that use TM to find unpublished sources. The initial search process was completed on January 10, 2022 and updated on September 21, 2022 in order to ensure no new important sources had been missed. This update did not reveal any new sources.

4 | RESULTS

Initially, we identified 1298 sources from the scientific databases and 29 sources from other databases, websites, etc. Figure 1 illustrates the selection process presented in a PRISMA-ScR flow diagram.

The results of the search strategy are presented in three ways: (1) In diagrammatic form which highlights the selection process of sources (Figure 1). (2) In tabular form which highlights central elements from the included sources (Table 4). (3) In a visual presentation of a conceptual mapping expressing a narrative summary of the patients' experiences with TM solutions (Figure 2).

4.1 | Characteristics of included sources

Our review included 13 sources published between 2004 and 2021, the majority ($n = 8$) during the period 2013–2021. The sources originated from different continents, countries and cultural areas: Australia ($N = 2$), the United States ($N = 2$), Canada ($N = 2$) and Scandinavian ($N = 7$). Eleven of the sources included were scientific in nature (qualitative methods ($N = 7$) and quantitative methods ($N = 4$)) and two were of general interest or non-scientific nature, presented with one case report and one interview in a newsletter. The included sources were published in 13 different journals/databases. The included sources are presented in Table 4.

4.2 | Conceptual mapping and narrative summary of patients' experiences

In the 13 sources included, we identified four main maps and two sub-maps reporting the most common experiences and views of the patients using TM solutions for the care and treatment of their DFUs. The sub-mapping was iteratively prepared during the process when we identified the main maps.

Overall, we found that the issues related to having a DFU were experienced as a big additional challenge day to day for patients living with DM. The patients stressed that on one hand, they were encouraged by the health professionals to live as normal a life as possible, but on the other hand, they had to spend a lot of energy focusing on changes and symptoms from their feet and/or DFUs. The patients also reported feelings of fear and anxiety, because they always had to live with the risk of repeated admittance to hospital, recurrence of the DFU and ultimately amputation.[23-28]

The four main maps with related sub-maps identified are listed below and illustrated in Figure 2, including the

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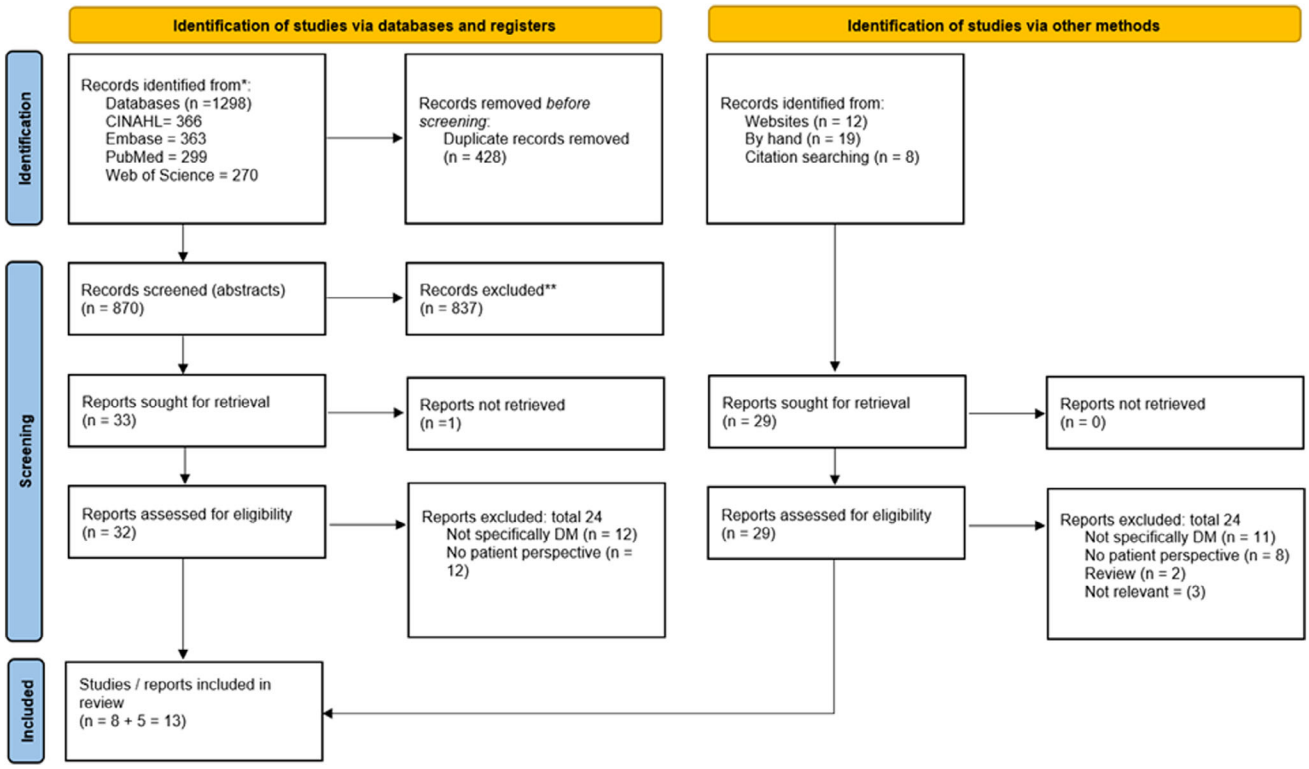


FIGURE 1 Selection process presented in a PRISMA-ScR flow diagram

individual numbers of sources referring to the issue of the map:

1. A whole human not merely a hole in a human
   a. The need-to-have and the nice-to-have
   b. My time and your time
2. Less of a burden on the family, the community and the environment
3. Competences and continuity of care are essential for high-quality care
4. The quality and modality of the technology

4.3 | Conceptual mapping and narrative summary

4.3.1 | Main Map 1: A whole human not merely a hole in a human

This main map representing most sources (12/13 sources) focuses on the crucially important point that the patients want to be viewed "as *a whole human not merely as a hole in a human"* by the health care professionals.[23-26,28-35] Clemensen et al report that the patients experienced the use of a TM solution moved the health care professionals' interest from solely the DFU to seeing the patient as a person. A patient expressed that "the good doctor was able to see me in my living room" and "somehow saw me as a whole person."[29] The patients experienced that the TM solution made them a part of the care, treatment and plan, and they felt empowered.

In the TM modalities where the patients were encouraged to report wound assessments, for example, weekly, using pictures, they experienced a greater feeling of responsibility and awareness of changes, regarding signs of infection or other changes involving the DFU.[26,29] Even when the community care nurse, the podiatrist, the endocrinologist and/or the GP participated, they experienced that they were equal partners and not just the human with a hole in the foot.[23,26,29,32-34]

By elaborating this map, we identified two sub-maps suggesting that the positive experiences and feelings of empowerment were related to other issues. According to the WHO, empowerment constitutes *"a process through which people gain greater control over decisions and actions affecting their health."*[36] We named these sub-maps "*The need-to-have and the nice-to-have*" (11/13 sources) and "*My time and your time*" (11/13 sources).

4.3.2 | Sub-map 1a: The need-to-have and the nice-to-have

The sub-map '*The need-to-have and the nice-to-have'* suggests that the TM solution gave the patients an experience of being able to minimise the stress of keeping track

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TABLE 4 Table illustrating the included sources in alphabetic order by the first author

Author, publication year, and country for the study	Title	Aim/objective	Publications type	Design	Main results/findings
Andersen, Signe Lindgård Steffen, Vibeke (2013). Denmark	Disease, normality and self-care: Diabetic foot ulcers and the logic of free choice.	To show how certain perceptions of DM as an easy-to-treat disease have consequences for the development and treatment of foot ulcers.	Scientific paper.	Qualitative study: Narrative interviews.	People with diabetic foot ulcers are a vulnerable group of patients with needs of both care and freedom of choice. TM is described as an opportunity to give patients peace of mind. Telemedicine contributes to a holistic practice in physical, psychological and practical circumstances when diagnosing and treating foot ulcers, thereby reducing the gap between hospital and home and increasing focus on holistic understanding.
Boodoo, C., et al (2007). Canada	Views of Patients on Using mHealth to Monitor and Prevent Diabetic Foot Ulcers: Qualitative Study.	To explore the views, attitudes and experiences of a diabetic patient population with or at risk of DFUs regarding technology, mHealth and the diabetic foot.	Scientific paper.	Qualitative study: In-depth interviews.	Patients expressed interest in two-way communication with HCPs that would allow for easier appointment scheduling, sharing of medical data, reduced number of visits and the use of alerts for when medical attention is required. Patients also identified conditions of the mHealth intervention. These included consideration of debilitating complications because of DM, such as retinopathy and decreased mobility; ease of use of the intervention; and implementation of virtual communities to support the continued use of the intervention.
Clemensen, J., et al (2005). Denmark	Telemedical treatment at home of diabetic foot ulcers.	To investigate the use of telemedicine to enable a visiting nurse (in the patient's home) to coordinate the treatment with experts (at the hospital).	Scientific paper.	Field studies were carried out at the outpatient clinic and by following visiting nurses. Five workshops and experiments were conducted in the laboratories, at the hospital, and joint events with hospital staff and patients. A pilot test with teleconsultations at home instead of visits to the hospital.	Both patients and relatives felt that they were a part of the conversation. The patient expresses satisfaction with the source of treatment and relief at not having to go to the hospital and spending much time waiting for transport and consultation.
Clemensen, J., et al (2007). Denmark	Telemedical teamwork between home and hospital: a synergetic triangle emerges.	To describe the effect on the clinical cooperation when introducing video consultations in patients' homes.	Conference abstract.	The study was conducted as a Participatory Design process containing workshops, field studies, clinical experiments and pilot tests involving participants from the Danish primary and secondary care sectors as well as patients and relatives. The video consultation set-up constitutes a new organisational way of working, described as "a new triangle" based on immediate inter-individual cooperation and teamwork.	In the triangle, competencies were combined which led to more holistic treatment and a more active role for the patient. Furthermore, a greater exchange of knowledge among all participants was observed, resulting in an upgrade of the competencies of especially the visiting nurse. The introduction of a real-time, online link between hospital and home constitutes the basis for simultaneous communication between all participants, potentially securing and enhancing the quality of treatment.

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TABLE 4 (Continued)

Author, publication year, and country for the study	Title	Aim/objective	Publications type	Design	Main results/findings
Devine, J. (2009). Australia	User satisfaction and experience with a telemedicine service for diabetic foot disease in an Australian rural community	To understand the experience of receiving medical management via telemedicine for rural patients with a diabetic "high-risk foot" diagnosis.	Scientific paper.	Qualitative study: Semi-structured face-to-face interviews.	The experience of the service was very positive for the patients as well as the carers. Initial responses when offered the service were positive and there was a strong common theme of frustration with the present care being provided. Being present at the time of the consultation was also not important to the patients and 100% said they would use this type of service again. The patients commented on the stress associated with travel to an urban area for medical care and cited the greatest benefit for them was the reduction of need to travel, as a result of the telemedicine.
Iversen, M.M., et al (2019). Norway	Effect of telemedicine intervention for diabetes-related foot ulcers on health, wellbeing, and quality of life: secondary outcomes from a cluster.	To compare self-reported health, wellbeing and QOL between patients receiving telemedical care and patients receiving standard care in an ulcer clinic.	Scientific paper.	Multi-centre cluster randomised controlled trial. 94 in TM group, 88 in control group. 42 clusters in 21 pairs. Patient-reported outcome measures were collected at baseline. Following measurements were used: Euro-Qol, HADS, PAID and NeuroQOL.	No significant effect or deterioration in patient-reported outcome measures during the study period.
Kim, H. M., et al (2004). USA	Patient attitudes towards a Web-based system for monitoring chronic wounds.	To evaluate patient attitudes towards a store-and-forward telemedicine system for monitoring chronic wounds.	Scientific paper.	A questionnaire developed for the study was administered to every participant during the first and third visits. 69 patients.	Patients were generally comfortable with the use of a telemedicine system to assess wound status. Less than 25% of the patients felt that using telemedicine at home would be worse than visiting their doctor in terms of the quality of care, even though about 85% of the patients felt that it was "important" or "very important" to be seen in person by a doctor for a pressure sore. Perhaps more importantly, less than 12% of the patients felt "somewhat or very unconfident" that their doctors would be able to know about the pressure sore via the telemedicine system instead of seeing the wound in person. These findings suggest that the ideal system of care, as perceived by patients, might be routine monitoring using the telemedicine system with occasional in-person visits to a physician, and that the value of the system may vary by wound type and setting.
Ploderer, B., et al (2018). Australia	Promoting Self-Care of Diabetic Foot Ulcers Through a Mobile Phone App: User-Centered Design and Evaluation.	To evaluate the usability and potential usefulness of a mobile phone app to engage people with DFUs in self-care.	Scientific paper.	Qualitative study: Semi-structured interviews.	MyFootCare shows promising features to engage people in DFU self-care. Most notably, ulcer size data are useful to monitor progress and engage people. However, more work is needed to improve the usability and accuracy of MyFootCare, for example by refining the process of taking and analysing photos of DFUs and removing unnecessary features.
Sandstrøm, H., et al (2006). Sweden	Telemedicine in a Multidisciplinary Diabetic Foot Care Service-A Pilot Study from Northern Sweden.	To test the feasibility of telemedicine in the management of DM-related foot problems in a rural area.	Conference abstract.	Case series: 10 patients with diabetic foot ulcers.	On a scale from 1 to 6 where 6 is very good, the patients scored their satisfaction with consultation via TM as 4.6.

(Continues)

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TABLE 4 (Continued)

Author, publication year, and country for the study	Title	Aim/objective	Publications type	Design	Main results/findings
Smith-Strøm, H., et al (2018). Norway	The Effect of Telemedicine Follow-up Care on Diabetes-Related Foot Ulcers: A Cluster-Randomised Controlled Non-inferiority Trial.	To evaluate if TM follow-up of DM-Related Foot Ulcers is inferior to standard outpatient care in respect of ulcer healing time and frequency of amputation and death.	Scientific paper.	A Cluster-Randomised Controlled Trial. 182 patients from 42 municipalities/ districts with diabetic foot ulcers were included. Patients' experiences measured by GS-PEQ at the end of the follow-up period.	1. No statistical difference in healing time between the two groups TM and standard outpatient care 2. No excess mortality in TM group 3. Less amputations in TM group 4. Equal number of consultations in both groups 5. Patient satisfaction equal in both groups 6. Patients in both groups felt secure and had confidence in the care that they received
Smith-Strøm, H., et al (2016). Norway	An integrated wound care pathway, supported by telemedicine, and competent wound management. Essential in follow up care of adults with diabetic foot ulcers	To uncover patients' experiences with telemedicine follow-up compared to traditional follow-up.	Scientific paper.	Qualitative study: individual semi-structured interviews. 24 participants recruited from an RCT.	Same degree of fear of amputation or new ulcers in the two groups • Patients lost confidence in the ulcer treatment, if they encountered lack of competencies and continuity of care • Telemedicine is an important supplement if used as intended • Need of access to specialist was only important, if they encountered lack of competencies and continuity of care
Suzuki, K. Clinical editor Panellists: Kim P.J, Tickner, A. (2020). USA	Where are We with Telemedicine and Wound Care?	To provide status on experiences with using telemedicine during the Covid-19 pandemic.	Panel discussion.		Dr Kim states that patients are fond of it, as it cuts down on travel time—but the technology can be challenging. Dr Tickner states that it reduces stress in the patients.
Kong, Ling Yuan, et al (2021). Canada	A 57-Year-Old Man with Type 1 Diabetes Mellitus and a Chronic Foot Ulcer Successfully Managed with a Remote Patient-Facing Wound Care Smartphone Application.	To provide a case report on the use of TM for a patient with a diabetic foot ulcer.	Case Report.	Case Report.	Three successive wounds were managed with a combination of in-person and telehealth visits complemented by the app. The patient found the app "educational and empowering."

of several appointments with different clinics, in different places and with different health care professionals, but also of remembering the important issues for each scheduled consultation.[24-26,28-35] When they participated in a TM consultation, the patients felt that they were taking part in a community meeting where every attendee had a shared responsibility for the outcome of the meeting. This feeling reduced the patients' feelings of stress related to being a patient.[23,25,26,30]

Within the sub-map *"The need-to-have and the nice-to-have,"* the patients also experienced that it was easier to get in contact with the health professionals when using a TM solution, especially because they got an immediate answer to their problem.

Our sources revealed that TM offered the patients a minimum of transport and time spent in relation to the care and treatment of their DFUs. A patient explained how the TM solution helps save time particularly due to less time spent on transport to hospital or the outpatient clinic.[23] This was also reported from the Australian and Canadian sources where patients living in distant rural areas had to take several days off to attend a consultation. Furthermore, the patients who lived in rural areas also reported the benefit of avoiding the stress related to driving in large unfamiliar cities looking for parking spaces, somewhere to eat and for toilet facilities.[25,26]

4.3.3 | Sub-map 1b: My time and your time

In the sources, the previously described feeling of empowerment[23,26,32-34,37] seems to give rise to critical reflection regarding *time* spent in connection with the TM solution.[23-26,28-32,34,35] According to the patients, the

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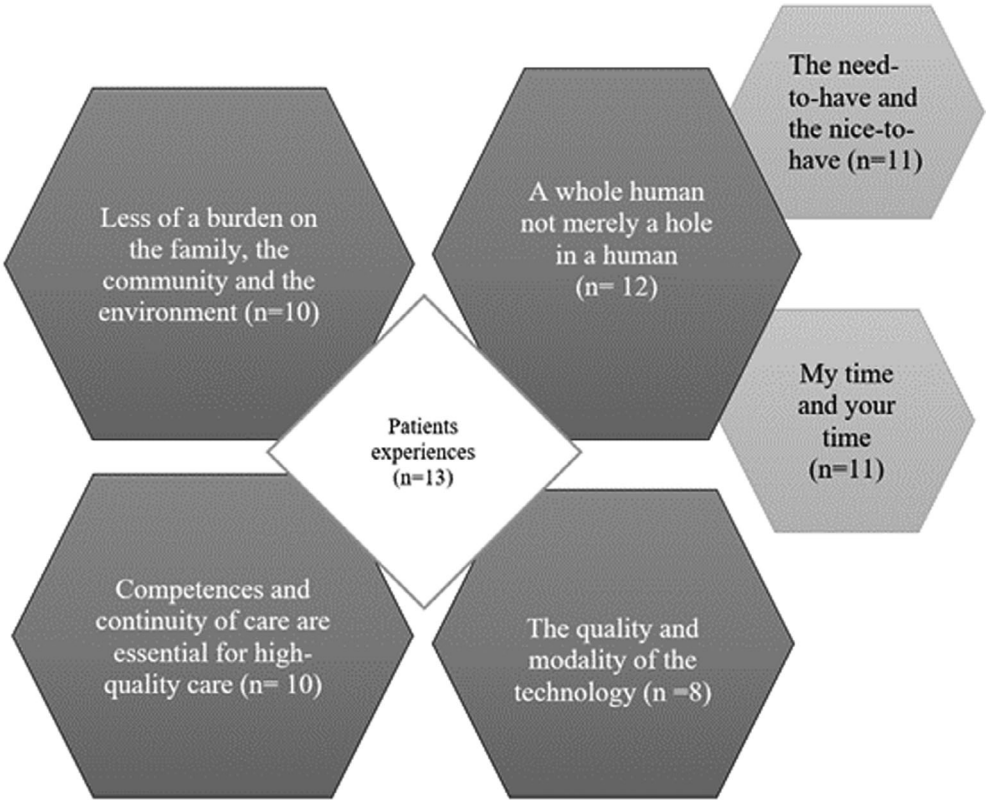


FIGURE 2 Illustration of the four main maps with their respective sub-maps

waiting time where they feared the outcome of the DFU consultation was a very stressful time independently of the setting.[23-25,28,32] The sources report that patients found it very important that the TM consultation or feedback on pictures kept to the agreed schedule.[23,38] If the health professionals did not keep to the agreed schedule, the patients would often blame the TM solution and be less eager to use the sustainable modality.[23,24,38] In addition, waiting for response in front of an app or a screen increased their sense of worry and was reported by the patients as a very different experience compared to waiting in a waiting room in a clinic or at a hospital.[23,24,29,31,38] Furthermore, the patients who used TM solutions also stressed the benefit of spending less time on transport or waiting in a waiting room. Thus, less time spent on transport or waiting increased patients' experiences of living an almost normal life without revealing their illness to others in their community or at their place of employment. For instance, a TM consultation could take place during a lunch break.[23,26,32]

4.3.4 | Main map 2: Being less of a burden on the family, the community and the environment

The second main map reports that a TM solution gave the patients the experience of being less of a burden on their families, the community and the environment (10/13 sources).[23-28,30-32,35] The patients did not want their relatives or friends to be obliged to help. The patients would rather have an equal and normal relationship with them all.[23,24,26] The TM solutions increased their experience of independence, which contributed to decreasing their overall guilty conscience of being a burden on their surroundings.

In addition, the patients had an awareness of TM as a pro-environmental issue as the burden of transport and traffic in the cities was reduced immediately.[24,25,32] Cost-related issues were also of importance to the patients coming from rural areas. TM offered an opportunity to cut the expenses related to a visit to the hospital or clinic. This was related to expenses regarding transport and possible overnight stays in the city where the hospitals were located.[25] Also, expenses related to the work place, where they had to miss a day's work, often with the result that they missed a day's pay.[23-25,28,30,31]

4.3.5 | Main map 3: Competences and continuity of care are essential for high-quality care

In terms of feeling safe, the patients attached equal importance to the health professionals' competencies regarding DFU care and treatment as to their

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technological know-how in using the TM modality (10/13 sources).[24-26,28-32,35,38] In situations where the patients experienced errors or incorrect treatment, they tended to lose confidence in the consultation and the health professionals' competencies and as a result, shy away from using the TM solution.[38] However, the opposite was also true: when a health professional demonstrated good or excellent competencies, the patient would use the functionalities of the TM solution more widely and more frequently.[25,32,35,38] Regarding their own understanding of technology, the patients often worried that they were not sufficiently technologically competent, leading to fears about missing important knowledge, information or plans for their treatment and care. In addition, patients with limited or no experience in the use of technology were very sceptical about using TM. This problem was reported especially in relation to elderly patients and patients characterised by, for example, social isolation, limited education and/or low social and economic status.[25,29,35]

4.3.6 | Main map 4: The quality and modality of the technology

With regard to the eight sources included in this main map, the functionality, the user-friendliness and the quality of the technology are cardinal points when it comes to patients feeling safe in using TM (8/13 sources).[23,24,26,28-31,34]

The patients described the TM solution as a tool to monitor the progress of their DFU, providing an immediate "view of the case" and with the aid of different functionalities, taking and frequently saving photos as time goes by. The patients regarded this presentation of the progress of their DFU as a support, motivating them to comply with restrictions and treatment. Thus, according to the sources, TM stimulated the patients' self-care. In addition, the TM solution was experienced as an easy way to contact the specialist when the initiated wound treatment was unsuccessful. As such, TM underpinned the patients' empowerment by facilitating actions affecting their health.

Technical challenges, such as slow or unstable internet, poor lighting, small screens and poor sound quality, did also affect whether or not the patients experienced the TM solution as beneficial for their wound treatment.[23,24,26,28-31,34] However, attention was also drawn to the consequences of other late complications related to DM, such as retinopathy or/and neuropathy which need to be taken into consideration when addressing practical aspects, such as screen colours, the size of the gadget and whether or not a touch screen is appropriate.[24]

4.4 | Strengths and limitations of the scoping review

The strengths of this scoping review are based on the extensive literature search extracting data from 13 sources. The sources were published in a span of 17 years, even though we set no limits regarding the year of publication. However, most of the sources ($n = 8$) were published within the last 7 years. The sources originated from three geographical areas: Australia, USA and Scandinavia. This may indicate that TM solutions are predominantly used in established Industrial countries and continents. All the same, the included studies represent different cultures where health care systems are structured and organised in different ways. The distribution may limit our results to describe only well-established industrial countries.

Furthermore, we found that TM was a very broad term, which included very different modalities, and that may influence the specificity of the findings of our review. We had no exclusion criteria concerning types of TM solutions.

5 | DISCUSSION

To the best of our knowledge, this is the first attempt to identify, examine and conceptually map the available literature on patients' experiences and views using TM solutions for remote monitoring, consulting and exchanging information between the patient, the homecare and hospital specialists regarding the treatment and care of their DFUs. The following recurring themes were identified during the review process.

5.1 | A person-centred holistic approach

It is of crucial importance for the patients with DFUs to be viewed "as *a whole human not merely as a hole in a human*" by the health care professionals. Our sources reveal that the patients with DFUs experienced TM as fulfilling this objective by supporting a more person-centred approach compared to the face-to-face visits in the clinic or at the hospital. In addition, the patients stressed that using a TM solution increased their experiences of feeling safe and empowered, as they felt they were included in the discussions of the development of their DFU care and treatment. The importance of an empowering approach to patients with DFUs was underlined by the effect on the patients' adherence to the initiated treatment and their level of self-care and

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participation in decision-making.[39] In addition, Fox points out that an increased focus on psychological support and a more committed approach from the health professionals seemed to contribute to patients being less fearful of further diabetes-related complications or ultimately amputations.[10]

We found that the empowerment stemmed from the fact that the TM solutions allowed the patients to live almost normal lives, often reducing the numbers of different contacts and consultations to fewer overall, though they tended to be slightly longer.

Moor et al stress that patients with DFUs often have a low quality of life, as the disease challenges their ability to create and sustain relationships, decreases their mobility and independence, and that the patients are often mentally affected by a troubling fear of amputation. However, Moore et al argue that a team-based approach in TM wound care have the potential to deal with such issues, because the team of health professionals have different competencies, approaches and focus on different aspects of the treatment and care.[7] According to our sources, the patients experienced that the TM solution offered a way for them to become an active part of the care, treatment and plan for their DFU. This is a more holistic approach, which also supported the patients' wishes to be seen as whole person and not merely a hole in a person.

Discussing a holistic approach, Foong et al pinpoint how health care professionals can provide better and more systematic documentation of the care provision when using TM, suggesting for example checklists as a method of generating systematic multidisciplinary reports.[40,41] The evidence-based guidelines produced by the International Working Group on the Diabetic Foot (IWGDF) include such checklists, and the IWGDF argues that a holistic and multidisciplinary approach will guide health professionals to investigate more than the ulcer when they apply a TM solution.[40-42]

To achieve a holistic approach, health professionals need to include the patient's perspective. Moor et al advocate the use of a shared checklist to gain a mutual understanding of the care which promotes seeing and including the whole person and thus improves the patient's quality of life.[7]

Moore et al present a reinterpretation of the ABCDE algorithm often used in critical care, providing an algorithm centred on TM assessment and care of—and in collaboration with—patients with DFUs (see Table 5).

However, the opposing point of view also exists, namely that a TM solution obstructs a multidisciplinary approach in situations where the solution only involved a physician.[40]

TABLE 5 ABCDE algorithm: critical care and wound care

Critical care	Wound care
Airways–assessing free and unobstructed airways	***A**ssessing patient and wound*
Breathing–assessing the respiration	***B**ring multidisciplinary team and informal carers*
Circulation–assessing the pulse, blood pressure and capillary response	***C**ontrol or teat systematically*
Disability–assessing the neurological status	***D**ecide appropriate treatment*
Exposure–assessing the exposure of the body	***E**valuate and reassess wound treatment and outcomes*

5.2 | Optimising time use

Our review seems to reveal a pattern where patients with DFUs felt empowered by making demands on active partnership concerning the planning and evaluation of their care and treatment. Furthermore, we found that the patients' perception of the benefit and quality of the TM solution was strongly correlated with the length of the waiting time. It is a well-known fact that time spent waiting for care providers influences the patient's experience of care. A study of waiting time for outpatients showed that the overall experience of satisfaction is negatively correlated with increased waiting time. In elaboration, the study showed that increased waiting time also affects outpatients' perceptions of information, instructions and the overall treatment provided by doctors and other caregivers.[43] Spechbach et al identify "*being forgotten*" as one of the strongest influencing factors in the perception of waiting times.[44] Furthermore, the patients' feeling of being abandoned is closely associated with lack of information about the delay.[45] Droit-Vole finds a distinct connection between time distortions and emotions and that the experience of time fluctuates according to the context and our emotional states. Time seems to fly when we are happy and to drag on when we are bored.[46]

We found that the patients with DFUs reported strong emotions, such as fear and anxiety regarding the risk of admittance, exacerbation of their ulcers and ultimately amputation. These perspectives in combination with the perspective on patients' perception of time could add to and improve our understanding of the patients feeling less empowered when they experience waiting times using a TM solution.

5.3 | Being less of a burden

We found that the patients often experienced ongoing controls, restrictions, admittance and consultations

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within several different health care facilities, and they did not want their relatives or friends to feel obliged to help. In relation to these tasks, the patients nevertheless had to ask relatives to do them favours regarding transport to the hospital, clinic or similar. The patients reported how such favours caused inconvenience to their relatives, both in relation to taking sick leave with the risk of impacting the family's finances, and in relation to, for instance, requesting a special diet in their workplace, at home or when they visited their family. We found that using TM gave the patients an experience of independence and an experience of being less of a burden on their family, the community and the environment.

Fox finds that lack of mobility combined with social isolation was a major challenge for the patient's self-understanding when undergoing care and treatment for DFU. Our findings suggest that these patients therefore often experienced altered family roles, reduced work opportunities and an overall interference in their daily lives. Such interference leading to stress and anxiety was closely connected with the patients' fear of lack of wound healing and risk of amputation. A patient expressed the hopelessness of not being able to participate in everyday activities and provide for the family, saying: "*that's how I feel: living without living.*"[10] Furthermore, the patients also talked about how the loss of livelihood caused significant financial hardship, they described being a financial burden on their families and the health care system. Such feelings were related to the patients' experiences of a change in social roles within the family as well as in the community, leading to a situation with self-imposed stress and a feeling of being a patient 24/7.[10]

Several studies indicated that the patients were very aware of how their disease and need for care and treatment could become a burden on the health care system.[2,9,13,40,47,48] The financial burden relating to a DFU treatment period may be anything up to 5.6 years of a patient's income and is almost independent of geography. This tendency is widely reported across the world, for example in China, Australian, India, Chile, Tanzania and Europe.[7] Thus, Armstrong et al stress that the new TM solutions offer patients with DFUs financially dominant strategies to manage their livelihoods.[2] TM was reported to improve social aspects, as patients were able to avoid unnecessary travelling and to be flexible in time and place, increasing the opportunities to go to work even during a period of treatment and care. In addition, TM might promote social engagement and acceptance and thereby promote self-management and empowerment for the patient.[40] The international consultancy firm Deloitte reports from its Sustainable Actions Index that shifts in people's attitudes regarding environmental sustainability influence behaviours, demands and choices regarding, for example, transport.[49]

The results of this scoping review are in alignment with these findings when it comes to being in favour of a TM solution: financial issues were of importance to the patients and the same goes for protecting the environment.

5.4 | The quality and modality of the technology

We found the quality and user-friendliness of the TM modality had a strong influence on the patients' experiences of using a TM solution. In addition, we found that an immediate "view of the case" was increasing the patients feeling of safety. The use of digital technology was seen as a mediator for patient feedback, which increases patient self-management and treatment adherence, and therefore have the potential to prevent recurrence of DFU.[1,9]

For the patients, the health professionals' competencies regarding wound care were of equal importance to their competences in managing the TM solutions.[14] However, the patients also had concerns about their own technology know-how and whether they had sufficient skills to use a TM solution. This was especially true of patients who did not use technology on a regular basis. Such patients were often elderly, or people who lacked financial resources or patients who did not use, for example, smartphones, tablets or computers.[40]

During the review of our sources, we found that technically related challenges, such as slow or unstable internet, poor lighting, small screens and poor sound quality, could be crucial in determining whether the patients experienced the TM solution as beneficial for their treatment and care. These findings were supported by other studies.[13] Furthermore, the technological equipment, the digital communication infrastructure and quality as well as the users' (both the patients and the health professionals) skills in mastering the technologies were described as crucial for patient satisfaction, patient outcome and for the health professionals' inclination to use the TM solution.[13]

Gomez-Trujillo et al argue that digital solutions and transformation relate to environmental sustainability but also to the sustainability of the health care system by improving customer-centred processes and flexibility and by reducing costs. They state that the people and organisations in charge need to be mindful of the potential of digitisation and balance the financial, environmental and social impacts of digital technologies to survive the digital revolution.[50]

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Finally, it has been argued that the challenge in using TM is primarily related to the overwhelming speed with which technology and informatics solutions evolve. Secondarily, the challenges are related to keeping the focus on the patient and the evidence-based treatment and care, avoiding the pitfalls of focusing solely on other qualities of TM, such as the functionalities, the innovative technological opportunities and the reduced costs.[8]

6 | CONCLUSION

We identified four main themes, describing the experiences of the patients using TM solutions for the care and treatment of their DFUs. From the patients' perspective, the idea of a TM solution was to promote holistic and person-centred care and treatment of their DFUs. Furthermore, patients reported how TM gave them an experience of being viewed as a whole person and not only as a hole in a person. Using a TM solution gave them the experience of having more control over their lives, being included in decisions, having fewer contacts, spending less time and often being able to lead an almost normal life. Overall, TM gave the patients a feeling of empowerment. Additionally, the TM solution gave the patients the feeling of being less of a burden on their families, the community and the environment. The patients experienced how they were able to sustain a job and secure their own and their family's livelihoods.

However, we found that the patients' views on or experiences with the quality and modality of the TM solutions could be a game changer for the use of a TM solution. Thus, the patients preferred to resume the traditional clinic visits, if the modality was viewed as functioning poorly, for example in regard to picture colour, speed of the internet, quality and size of the face-to-face screen or speed of response.

In conclusion, we recommend further investigation into a more specific evaluation of the use of TM solutions among patients with DFUs. We suggest investigating "*What works for whom and under what circumstances*" from the perspectives of both the patient and caregiver in the primary and secondary health care sectors. We recommend research performed across the two sectors with particular focus on person-centred care, multidisciplinary teamwork, organisational changes and environmental sustainability in the care and treatment of—and in collaboration with—patients with DFUs.

ACKNOWLEDGEMENTS

No acknowledgement.

CONFLICT OF INTEREST

We have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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How to cite this article: Søndergaard SF, Vestergaard EG, Andersen AB, Kolbæk R, Dahl M, Høgh A. How patients with diabetic foot ulcers experience telemedicine solutions: A scoping review. *Int Wound J*. 2022;1-15. doi:10.1111/iwj.14026

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